Toward a four-dimensional Yang–Mills construction

Notebook 15: a uniform nonlinear constrained Wilson minimum

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Abstract

We assemble a nonlinear small-field minimizer construction for the bare Wilson action on homogeneous finite periodic regulators. The constants and coarse-data analyticity radius are uniform in the refinement and volume. The proof requires an equal-exponent constrained Schauder estimate, an exact Wilson force factorization with removable mesh factors, and a two-Hilbert-direction bound for the nonlinear constraint curvature. These yield a strict minimum in a fixed flat gauge slice for the corrected average. A constraint-preserving gauge flow and a separate transfer to ordinary group averages then give a minimum, modulo gauge, among sufficiently small global physical C1-alpha potentials with prescribed ordinary coarse average. This is a finite-regulator classical ingredient for a constructive program. It does not yet construct an interacting continuum quantum theory.

Status and provenance. Research proofs subjected to separate internal mathematical audits; no external certification or novelty claim. Source matching refers to the deposited transcriptions at commit `e392ef45013a581664bab76b90867170b30fe10e` of Yang–Mills-LLM-experiment. Original CMP98 and CMP99b pages have not all been independently recovered. The action and physical pairing match the deposited four-dimensional bare Wilson conventions; their source-coordinate Hessian is distinguished explicitly from the ordinary-potential Hessian used in the nonlinear argument.

1 The result and the remaining construction problem

Fix a matrix group $SU(N_c)$, normalized trace $\tau = \mathrm{Tr}\,/N_c$, an odd block ratio $L \geq 3$, and $0 < \alpha < 1$. Fine spacing is $\eta = L^{-j}$ and the retained coarse spacing is one. The regulators are compatible finite periodic lattices. Potentials are Hermitian traceless on the real locus and are complexified for analytic estimates. The complete-field norm N_α controls the potential, its physical first differences and their α-Hölder seminorms. Transposes always use physical sums. Fixed collar enlargements and orientation multiplicities are included in the constants, which may depend on L, N_c, α.

The final theorem is Theorem 6.1. In words, for sufficiently small arbitrary coarse data B there is an analytic small fine field with the exact ordinary homogeneous constraint $M^{(j)}(U) = e^{iB}$, and its action is minimal among all fields in a fixed small N_α ball satisfying that constraint. Equality within the ball means gauge equivalence. The corrected representative also obeys the sharper expansion $A_*(B) = HB + O(\|B\|_\infty^2)$ in N_α, where H is the flat minimizing extension. All these bounds are uniform in the mesh and the compatible periodic volume.

There are three distinct operations in the proof. The flat Green operator solves the nonlinear stationarity equation in a fixed gauge slice. A genuine gauge flow covers nearby fields with the *same corrected constraint*. Finally, two explicit gauge constructions relate the corrected and ordinary constraint sets. Neither last step is a consequence of fixed-slice convexity alone. The ordinary representative need not be in the same flat gauge slice as the corrected one.

The flat operator identities, canceled Fourier aliases, and physical exponential locality used below were derived in Notebooks 8–12. The exact nonlinear constraint chart is from Notebook 13. Notebook 14 supplies the ordinary/corrected gauge relation and reference-coordinate comparison. This document contains the additional estimates that close the nonlinear minimum argument. Its symbols are restated at their use, but the earlier flat alias estimates remain explicit prerequisites.

This closes a concrete classical small-field step. It leaves open extension to local gauges and mixed retained histories, control of large fields, integration of interacting fluctuations, running-coupling estimates, and a nontrivial continuum observable hierarchy. Euclidean invariance, positivity, and the physical mass gap remain separate obligations. The gap in the constrained flat auxiliary problem is not a physical Yang–Mills mass gap.

2 The constrained flat inverse preserves the Hölder exponent

We now establish the constrained Green estimate needed for the nonlinear fixed-gauge argument. The estimate concerns the homogeneous flat operators. It is not a nonflat Schauder theorem. All fine sums use physical measure η^d; transposes below use that measure and are bilinear after complexification.

2.1 Definitions and exact constraints

Set

$$K = d^T d, \qquad \mathcal{L} = D^T D, \qquad \mathcal{A}_0 = K + DRD^T + Q^T a_{\mathrm{pen}} Q, \qquad G_0 = \mathcal{A}_0^{-1}, \tag{1}$$

where D is the scalar fine gradient, d the bond curl, and $Q = Q_M^{\mathrm{st}}$ the flat bond average. The one-form penalty is fixed in a compact positive interval. Define

$$q_{\mathrm{aux}} = (QG_0Q^T)^{-1}, \qquad H = G_0 Q^T q_{\mathrm{aux}}, \qquad \Gamma = G_0 - H q_{\mathrm{aux}}^{-1} H^T. \tag{2}$$

The matched flat extension satisfies $QH = I$ and $RD^T H = 0$. With the scalar source projection and its constrained biharmonic inverse, write

$$Z = G'RG', \qquad \Lambda = Z\mathcal{L}D^T = G'RD^T, \qquad \Pi = I - D\Lambda. \tag{3}$$

The established flat identities give

$$Q_0 Z = 0, \quad \mathcal{L}Z\mathcal{L} = R, \quad Z\mathcal{L}^2 Z = Z, \quad G_0 \Lambda^T = DZ. \tag{4}$$

Since $\Lambda H = 0$, they imply $\Gamma\Lambda^T = DZ$, $\Lambda\Gamma = ZD^T$, and $\Lambda\Gamma\Lambda^T = Z$. Expanding the two projection factors therefore proves

$$\boxed{\mathcal{C} = \Pi\Gamma\Pi^T = \Gamma - DZD^T}. \tag{5}$$

The cochain relation $QD = D_c Q_0$ gives $Q\Pi = Q$. Also $RD^T\Pi = RD^T - R\mathcal{L}Z\mathcal{L}D^T = 0$. Consequently

$$Q\mathcal{C} = 0, \qquad RD^T\mathcal{C} = 0, \qquad \mathcal{C}Q^T = 0. \tag{6}$$

The last identity uses symmetry of $\mathcal{C}$. Since $\Gamma\mathcal{A}_0 = I - HQ$, any $\xi \in \mathcal{X} := \ker Q \cap \ker(RD^T)$ satisfies $\Gamma K\xi = \xi$. The identity $D^T K = 0$ then gives

$$\mathcal{C}K\xi = \xi \quad (\xi \in \mathcal{X}), \qquad \mathcal{C}KH = 0. \tag{7}$$

For the second assertion, use $KH = Q^T(q_{\mathrm{aux}} - a_{\mathrm{pen}} I)$ and (6). On real regulators $\mathcal{C} = \Pi\Gamma\Pi^T$ is nonnegative and is exactly the inverse of the Maxwell form restricted to $\mathcal{X}$: its range lies in $\mathcal{X}$, and $\langle \xi, K\mathcal{C}f\rangle = \langle \xi, f\rangle$ for every $\xi \in \mathcal{X}$. The restricted form is coercive because there it equals the positive form $\mathcal{A}_0$.

2.2 Uniform statement

Let $S_{\mathrm{jet}} = (I, \nabla)$ contain every ordinary fine first difference of bond components. Finite component selections, shifts and curl contractions are allowed. For a fixed $0 < \alpha < 1$, use complete-field Hölder norms on fixed unit-cell collars and admissible positive weights satisfying $\varpi(n)/\varpi(m) \le e^{\kappa d(n,m)}$. The norm $N_{\alpha,\varpi}$ contains the field supremum, first-difference supremum, and the α-Hölder seminorm of the first differences. The input norm $\|f\|_{\alpha,\varpi}$ contains the weighted supremum and the collar α-Hölder seminorm.

Theorem 2.1. *For a sufficiently small fixed spatial exponent $\kappa > 0$, uniformly in refinement and compatible periodic volume,*

$$N_{\alpha,\varpi}(\mathcal{C}f) \le C_\alpha \|f\|_{\infty,\varpi}, \tag{8}$$

$$N_{\alpha,\varpi}(\mathcal{C}S^T_{\mathrm{jet}}(f_0, f_1)) \le C_\alpha(\|f_0\|_{\infty,\varpi} + \|f_1\|_{\alpha,\varpi}). \tag{9}$$

The same estimates hold on the full lattice. Thus the derivative datum has the same Hölder exponent as the output gradient. No assertion is made at either endpoint $\alpha = 0, 1$.

2.3 The extension subtraction in Gamma

The established flat equal-exponent estimate gives $N_{\alpha,\varpi}(G_0\nabla^T f) \le C_\alpha \|f\|_{\alpha,\varpi}$, and the bounded-forcing estimate gives $G_0 : \ell^\infty \to N_\alpha$. The localized extension satisfies

$$\|H(\cdot, b)\|_{C^{1,\alpha}(\text{collar } n)} \le C_\alpha e^{-\rho d(n,b)}. \tag{10}$$

Moreover $q^{-1}_{\mathrm{aux}} = QG_0Q^T$ has an exponentially summable coarse kernel. Hence

$$\Gamma\nabla^T = G_0\nabla^T - Hq^{-1}_{\mathrm{aux}}(\nabla H)^T \tag{11}$$

maps C^α_ϖ to $N_{\alpha,\varpi}$ uniformly. In fact its subtracted term accepts bounded data: the physical transpose row is an integral over cells of volume one, so (10) bounds it without any power of the fine-cell dimension. The weighted exponential kernels can then be convolved. The undifferentiated argument also proves $\Gamma : \ell^\infty_\varpi \to N_{\alpha,\varpi}$.

2.4 A massive biharmonic comparison

It remains to estimate $DZD^T\nabla^T$. Put $Z_m = (\mathcal{L} + 1)^{-2}$. We first prove the stronger remainder bound

$$N_{\alpha,\varpi}(D(Z - Z_m)D^T\nabla^T f) \le C_\alpha \|f\|_{\infty,\varpi}. \tag{12}$$

The canceled scalar alias formula is

$$Z_{00} = R_{s,2}/D_{s,2}, \qquad Z_{0s} = -\frac{a_0^- a_s}{P_s^2 D_{s,2}}, \qquad Z_{r0} = -\frac{a_r^- a_0}{P_r^2 D_{s,2}},$$
$$Z_{rs} = \frac{\delta_{rs}}{P_r^2} - \frac{P_0^2 a_r^- a_s}{P_r^2 P_s^2 D_{s,2}}, \qquad r, s \ne 0, \tag{13}$$

where $D_{s,2} = a_0 a_0^- + P_0^2 \sum_{r\ne 0} a_r a_r^-/P_r^2$. These expressions are uniformly holomorphic on a common enlarged tube, with all displayed canceled denominators bounded away from zero. A fixed further shrinking does the same for $P_0 + 1$; the high massive denominators are already gapped.

The high diagonal difference is exactly

$$P_r^{-2} - (P_r + 1)^{-2} = \int_0^\infty t(1 - e^{-t})e^{-tP_r}\, dt, \qquad r \ne 0. \tag{14}$$

There are three external differences in $D(Z - Z_m)D^T\nabla^T$, and at most one further output difference and an α-increment. The small-time absolute row bound is therefore

$$t(1 - e^{-t})t^{-(4+\alpha)/2} \le t^{-\alpha/2}, \qquad 0 < t \le 1, \tag{15}$$

which is integrable. The established complex heat-image bounds apply through these fixed derivative orders by semigroup splitting. The removed central mode is bounded at small times, and the high spectral gap controls large times.

For the rank terms, the high output vector a_r^-/P_r^2 is multiplied by at most $C(1 + |r|)^{2+\alpha}$. With $w(r) = \prod_\nu (1 + |r_\nu|)^{-1}$, its absolute sum is bounded by

$$C \sum_{j \ge 0} (1 + j)^{d-1} 2^{-j(2-\alpha)} < \infty.$$

The two input differences leave a row bounded by $C|a_s|(1 + |s|)^{-2}$, whose alias ℓ^2 norm is uniform. Central terms are bounded. Physical-cell L^2 is dominated by the supremum without a dimension factor. Reconstruction, periodic gluing of the complete operator, and contour shifting give the exponentially local version of (12). Only the reconstructed operator is periodized; separated central chart terms need not individually be periodic.

For the massive part use

$$Z_m = \int_0^\infty t e^{-t} e^{-t\mathcal{L}}\, dt. \tag{16}$$

The output gradient of $DZ_m D^T \nabla^T f$ has four fine differences of the heat kernel. The required next difference has five. For $k = 4, 5$ the weighted moment estimate is

$$\sum_v e^{\kappa|v|_1} |v|_1^\alpha |\nabla_\eta^k p_t^\eta(v)| \le C_{k,\alpha} t^{(\alpha-k)/2}, \qquad 0 < t \le 1. \tag{17}$$

For $t \ge \eta^2$, split into k heat factors and use the weighted first-difference estimate at exponential scale $\kappa + t^{-1/2}$; the extra exponential pays for the moment. For $t < \eta^2$, expand the fixed k differences into shifts of size at most $k\eta$. The dimensionless walk has time below one, giving $C\eta^{\alpha-k} \le Ct^{(\alpha-k)/2}$. Thus no sub-mesh time range is omitted.

Every derivative row has sum zero. Subtracting the value of f at the output point and using the complete weighted Hölder increment bound gives

$$\begin{aligned} \left\| t\nabla^4 e^{-t\mathcal{L}} f \right\|_{\infty,\varpi} &\le Ct^{-1+\alpha/2} \|f\|_{\alpha,\varpi}, \\ \left\| t\nabla^5 e^{-t\mathcal{L}} f \right\|_{\infty,\varpi} &\le Ct^{-3/2+\alpha/2} \|f\|_{\alpha,\varpi}. \end{aligned} \tag{18}$$

For output separation $\eta \le r \le 1$, split at $t = r^2$: below it use twice the first bound, and above it use r times the second along a fine path in an enlarged fixed collar. The gradient difference, divided by the input norm, is at most

$$C \int_0^{r^2} t^{-1+\alpha/2}\, dt + Cr \int_{r^2}^1 t^{-3/2+\alpha/2}\, dt + Cr \le C\left(\frac{2}{\alpha} + \frac{2}{1-\alpha} + 1\right) r^\alpha. \tag{19}$$

The last term is the massive large-time tail, with a sufficiently small fixed exponential weight. Larger separations use the supremum bound. The gradient supremum is integrable for $\alpha > 0$; the field supremum has absolute small-time budget $t\, t^{-3/2} = t^{-1/2}$. Therefore

$$N_{\alpha,\varpi}(DZ_m D^T \nabla^T f) \le C_\alpha \|f\|_{\alpha,\varpi}. \tag{20}$$

Combining (12) and (20) proves the differentiated subtraction estimate. Without the last input derivative, the absolute budget is $t\, t^{-(3+\alpha)/2}$, integrable for $\alpha < 1$; this gives $DZD^T : \ell^\infty_\varpi \to N_{\alpha,\varpi}$. Equation (5) and (11) now prove Theorem 2.1.

2.5 Full-lattice identities and the application boundary

The massive estimates use complete fields and global heat-row cancellation. The remainder estimates use exponentially local pointwise cell kernels with input derivatives already on the kernel. No sharp restriction of a Hölder field to one cell is used. Exact periodization preserves the estimates, including small compatible tori and their incidence multiplicities.

At each fixed spacing the identities above extend from Hilbert fields to bounded or admissibly weighted complete fields by absolute summability of the operator products. In particular $R = \mathcal{L}Z\mathcal{L}$ is exponentially summable; fixed-mesh equivalence of cell norms suffices for this identity argument. It is not necessary to call the gauge projection local. For $\xi \in \mathcal{X} \cap N_\alpha$, $K\xi$ can be read as a first-divergence datum with Hölder coefficients, and (7) agrees with that interpretation.

This supplies the bounded-forcing and equal-exponent jet-forcing Green inputs for the fixed-flat-gauge nonlinear stationarity argument. The nonlinear action force, nonlinear constraint derivatives and strict-minimum argument are separate ingredients. No moving-gauge nonflat Schauder estimate, mixed retained-history result or interacting continuum limit is asserted here.

3 A uniform nonlinear Wilson force in physical coordinates

We work first on finite compatible periodic regulators, with fine spacing $0 < \eta \le 1$, physical measure η^d and normalized trace $\tau = \mathrm{Tr}\,/N_c$. The Lie algebra consists of Hermitian traceless matrices and is complexified for analytic arguments. All transposes use the fixed bilinear trace pairing. Constants are independent of refinement and volume. On the infinite lattice the formulas concern local variations; an arbitrary bounded field need not have finite total action.

In dimension four the action, trace and physical pairing below match the deposited conventions: Raw I:454–467 sums the normalized Wilson density once per unoriented plaquette, I:474–479 places the coupling outside that action, and I:12756–12760 specifies the physical bilinear pairing. The source fluctuation chart is explicit at I:15643–15648. Here we differentiate the same action in the ordinary global coordinate $U = e^{i\eta A}$; its Hessian at nonzero A is not silently identified with a differently based left-exponential Hessian. The original journal texts have not all been independently reverified.

3.1 Exact plaquette density and its removable scale factor

For $p = (x; \mu, \nu)$, $\mu < \nu$, define the physical jet

$$(\mathsf{S}_\eta A)_p = (a, b, c, e) = \big(A_\mu(x), A_\nu(x), \nabla_{\eta,\mu} A_\nu(x), \nabla_{\eta,\nu} A_\mu(x)\big). \tag{21}$$

Here $\nabla_{\eta,\rho} f(x) = \eta^{-1}(f(x + \eta e_\rho) - f(x))$. The plaquette holonomy is exactly

$$P_\eta(a, b, c, e) = e^{i\eta a} e^{i\eta b + i\eta^2 c} e^{-i\eta a - i\eta^2 e} e^{-i\eta b}. \tag{22}$$

This merely rewrites shifted link potentials; it is not a Taylor approximation. In potential coordinates set

$$\mathcal{W}_\eta(A) := \mathcal{S}_\eta(e^{i\eta A}) = \eta^{d-4} \sum_p \left[1 - \frac{1}{2}\tau(P_p + P_p^{-1})\right]. \tag{23}$$

In $d = 4$ the prefactor is one, as in the deposit. For other fixed d the displayed normalization defines the model used in this local calculation.

Lemma 3.1. *The matrix*

$$\mathfrak{F}_\eta(z) = \frac{P_\eta(z) - I}{\eta^2}$$

extends jointly holomorphically through $\eta = 0$. The density

$$L_\eta(z) = -\frac{1}{2}\tau(\mathfrak{F}_\eta(z)^2 P_\eta(z)^{-1}) \tag{24}$$

has uniformly bounded jet derivatives of every fixed order for $\eta \in [0,1]$ on a fixed smaller complex jet ball. Its quadratic Taylor part at $z = 0$ is exactly

$$L_2(z) = \frac{1}{2}\tau((c-e)^2), \tag{25}$$

independently of η. Moreover

$$\mathcal{W}_\eta(A) = \eta^d \sum_p L_\eta((\mathsf{S}_\eta A)_p). \tag{26}$$

Proof. The matrix product (22) is entire in η and the finitely many complex jet coordinates. For every z, $P_0(z) = I$ and $\partial_\eta P_\eta(z)|_{\eta=0} = i(a+b-a-b) = 0$. Thus the extension is explicitly

$$\mathfrak{F}_\eta(z) = \int_0^1 (1-s)\,\partial_\eta^2 P_{s\eta}(z)\,ds, \tag{27}$$

where the derivative is with respect to the scalar argument before evaluation at $s\eta$. The inverse P_η^{-1} is the reverse product of inverse exponentials and is entire. On $|\eta| \le 2$ and a fixed bounded jet polydisc these expressions are bounded. Cauchy estimates in the jet coordinates give the asserted uniform derivative bounds on a smaller polydisc.

At $z = 0$, $\mathfrak{F}_\eta = 0$, and direct differentiation gives

$$D_z P_\eta(0)[z] = i\eta^2(c-e), \qquad D_z\mathfrak{F}_\eta(0)[z] = i(c-e). \tag{28}$$

Equation (25) follows from (24). In particular $\mathfrak{F}_0(z) = i(c-e) - [a,b]$; the commutator is quadratic in the jet and does not change the quadratic density. Finally, the exact algebraic identity

$$P + P^{-1} - 2I = (P-I)^2 P^{-1}$$

proves (26). All inverse mesh factors cancel before a force derivative is taken. □

3.2 The exact force and its local nonlinear coefficient

Put $R_\eta = L_\eta - L_2$ and define the jet gradient f_η by the fixed bilinear trace pairing:

$$DR_\eta(z)[w] = \sum_{\ell \in \{a,b,c,e\}} \tau(f_{\eta,\ell}(z) w_\ell). \tag{29}$$

Projection onto the fixed traceless space is included. On a fixed small complex jet ball, uniformly in η,

$$\begin{gathered} f_\eta(0) = Df_\eta(0) = 0, \qquad \|f_\eta(z)\| \le C\,\|z\|^2\,, \qquad \|Df_\eta(z)\| \le C\,\|z\|\,, \\ \left\|D^2 f_\eta(z)\right\| + \left\|D^3 f_\eta(z)\right\| \le C. \end{gathered} \tag{30}$$

These follow from Lemma 3.1 and Taylor's formula, since R_η begins at degree three.

Let d be the ordinary fine curl, so $(dA)_p = c - e$, and let $K = d^T d$. Define $E_\eta(A)$ by $D\mathcal{W}_\eta(A)[H] = \langle E_\eta(A), H\rangle_\eta$. Then

$$E_\eta(A) = KA + \mathsf{S}_\eta^T \mathcal{N}_\eta(A), \qquad \mathcal{N}_\eta(A)_p = f_\eta((\mathsf{S}_\eta A)_p). \tag{31}$$

For a four-slot plaquette field q, the transpose is explicitly

$$\begin{aligned}(\mathsf{S}_\eta^T q)_\lambda(x) = &\sum_{\nu>\lambda}\left[q^a_{\lambda\nu}(x) + \nabla^T_{\eta,\nu} q^e_{\lambda\nu}(x)\right] \\ &+ \sum_{\mu<\lambda}\left[q^b_{\mu\lambda}(x) + \nabla^T_{\eta,\mu} q^c_{\mu\lambda}(x)\right],\end{aligned} \tag{32}$$

where $\nabla^T_{\eta,\rho} q(x) = \eta^{-1}(q(x-\eta e_\rho) - q(x))$. Every slot has the same physical measure η^d. At quadratic order the c, e coefficients are $c - e, e - c$, confirming $K = d^T d$ with coefficient one. The derivative in (32) stays outside the nonlinear coefficient; a uniform pointwise bound on the unprocessed nonlinear force is not asserted.

3.3 Physical Hölder norms without a mesh loss

Fix $0 < \alpha < 1$. All fields are complete fields, with componentwise comparisons at translated sites. Define

$$\begin{aligned}N(A) &= \|A\|_\infty + \max_\rho \|\nabla_{\eta,\rho} A\|_\infty, \\ N_\alpha(A) &= N(A) + \max_\rho [\nabla_{\eta,\rho} A]_\alpha.\end{aligned} \tag{33}$$

The seminorms use physical distances in fixed enlarged unit-cell collars. Suprema supply longer-distance increments. This is equivalent to the complete-field $C^{1,\alpha}$ norm used in the minimizer argument, with constants independent of η.

For two points at physical distance at most a fixed R, telescoping along an axis-aligned path gives

$$\frac{|A(x) - A(y)|}{|x-y|^\alpha} \le \sqrt{d}\, R^{1-\alpha} \max_\rho \|\nabla_{\eta,\rho} A\|_\infty. \tag{34}$$

At nearest-neighbor separation the factor is $\eta^{1-\alpha}$, not $\eta^{-\alpha}$. The derivative jet slots already have their Hölder seminorms in (33). Consequently

$$\|\mathsf{S}_\eta A\|_\infty \le C_J N(A), \qquad \|\mathsf{S}_\eta A\|_{C^\alpha} \le C_J N_\alpha(A). \tag{35}$$

No second-difference supremum is needed.

For a positive cell weight ϖ with $\varpi(n)/\varpi(m) \le e^{\kappa d(n,m)}$, weight each complete local collar norm by $\varpi(n)$. Denote the resulting norms by C^α_ϖ, N_ϖ and $N_{\alpha,\varpi}$. Neighboring fixed collars have comparable weights. The same telescoping argument gives

$$\|\mathsf{S}_\eta H\|_{C^\alpha_\varpi} \le C_{J,\kappa} N_{\alpha,\varpi}(H), \qquad \|\mathsf{S}_\eta H\|_{\infty,\varpi} \le C_{J,\kappa} N_\varpi(H). \tag{36}$$

No derivative is taken of a weight and no field is cut sharply into cells. Periodic lifting gives the same bounds on compatible tori.

Theorem 3.2. *On a fixed small complex N_α ball, $\mathcal{N}_\eta$ is holomorphic and, uniformly in refinement and volume,*

$$\|\mathcal{N}_\eta(A)\|_{C^\alpha_\varpi} \le C N(A) N_{\alpha,\varpi}(A), \tag{37}$$

$$\|\mathcal{N}_\eta(A) - \mathcal{N}_\eta(B)\|_{C^\alpha_\varpi} \le C(N_\alpha(A) + N_\alpha(B)) N_{\alpha,\varpi}(A - B), \tag{38}$$

$$\|D\mathcal{N}_\eta(A)[H]\|_{C^\alpha_\varpi} \le C\left[N(A) N_{\alpha,\varpi}(H) + N_\alpha(A) N_\varpi(H)\right]. \tag{39}$$

Weighted estimates require only that their right sides be finite. The base fields are in the unweighted small ball; one may put the weight on a direction or difference.

Proof. On each collar the product identity gives

$$[uv]_\alpha \le \|u\|_\infty [v]_\alpha + [u]_\alpha \|v\|_\infty .$$

It remains valid with the weight on either factor. Write $z = \mathsf{S}_\eta A$. The supremum bound in (30) and integration of Df_η along the segment from $z(x)$ to $z(y)$ give

$$|f_\eta(z(x)) - f_\eta(z(y))| \le C \|z\|_\infty |z(x) - z(y)|.$$

Together with (35) and (36), this proves (37).

The exact derivative is $Df_\eta(z)\mathsf{S}_\eta H$. Its coefficient has supremum at most $C\|z\|_\infty$ and Hölder seminorm at most $C[z]_\alpha$, since $D^2 f_\eta$ is bounded. The product estimate proves (39). Integrating it along $B + s(A - B)$ proves (38). In particular the unweighted local Lipschitz constant on a radius-r ball is at most Cr.

The uniform local Taylor series for f_η starts in degree two. The Hölder product inequality and (35) make its substitution converge absolutely on a fixed smaller N_α ball. This proves holomorphy in the complete Banach norms. Higher derivatives follow from the same series or Cauchy's formula. With one weighted direction, use the weighted product inequality term by term. If a weight is not bounded below, the weighted statement is read on the intersection with the unweighted small ball; no embedding into that ball is presumed. □

For example the same proof gives

$$\left\| D^2 \mathcal{N}_\eta(A)[H, K] \right\|_{C^\alpha_\varpi} \le C(1 + N_\alpha(A)) N_\alpha(H) N_{\alpha,\varpi}(K), \tag{40}$$

with the weighted direction exchangeable by symmetry. Uniform bounds on a complex ball give factorial Cauchy bounds on every smaller ball. All these statements are about nonlinear coefficients before the exterior transpose in (31).

3.4 Energy Hessian and the minimizer interface

Differentiating (31) once gives

$$DE_\eta(A) = K + \mathsf{S}_\eta^T Df_\eta(\mathsf{S}_\eta A)\mathsf{S}_\eta. \tag{41}$$

The middle coefficient is symmetric for the fixed bilinear transpose, is local in the plaquette jet, and has supremum at most $CN(A)$. Thus for physical Hilbert directions H, K,

$$\left| D^2 \mathcal{W}_\eta(A)[H, K] - \langle dH, dK \rangle_\eta \right| \le CN(A) \|\mathsf{S}_\eta H\|_{2,\eta} \|\mathsf{S}_\eta K\|_{2,\eta} . \tag{42}$$

This energy statement needs small $N(A)$, not a second background derivative. It is an ordinary-potential Hessian identity for the exactly specified Wilson action.

For the fixed-gauge minimizer, the equal-exponent estimate for its selected flat constrained inverse $\mathcal{C}$,

$$\mathcal{C}\mathsf{S}_\eta^T : C^\alpha \longrightarrow N_\alpha,$$

combines directly with Theorem 3.2. The resulting nonlinear map is quadratic and has Lipschitz constant $O(r)$ on a radius-r ball, also for a single weighted direction. The constrained Green estimate is proved separately; a positive loss between input and output exponents would not suffice for this step. No additional fine derivative of $\mathcal{N}_\eta(A)$ or inverse power of η is used in the contraction.

4 A uniform local constrained Wilson minimum on the flat gauge slice

Fix an odd $L \geq 3$, $M = L^j$, $\eta = M^{-1}$, final coarse scale one, and $0 < \alpha < 1$. The primary statement is on finite compatible periodic regulators. Constants may depend on L, dimension, the color norm, and α, but not on j or periodic volume. All transposes use the physical fine pairing $\eta^d \sum \tau(AB)$ and the corresponding coarse pairing; at complex fields they are bilinear transposes. In this module $\mathcal{S}(A)$ means the ordinary-potential action $\mathcal{W}_\eta(A) = \mathcal{S}_\eta(e^{i\eta A})$ of the force section, and $S_{\text{jet}} = \mathsf{S}_\eta$. Write the average in its fixed-reference potential coordinate as

$$\mathcal{F}(A) = QA + \mathcal{R}(A), \qquad Q = Q_M^{\text{st}}, \qquad \mathcal{R}(0) = D\mathcal{R}(0) = 0. \tag{43}$$

The symbol R without calligraphic font continues to denote the fixed flat scalar gauge projection. In particular, R and $\mathcal{R}$ are different objects.

4.1 Two uniform estimates for the nonlinear constraint

The previously established nonlinear averaging ball gives a one-step dimensionless corrected map F_L and physical map

$$\mathcal{F}_h(A) = (Lh)^{-1} F_L(hA) = Q_h A + \mathcal{R}_h(A). \tag{44}$$

Let m_L bound the number of input bonds read by one output, and N_L the number of outputs reading one input. On a fixed local complex neighborhood choose K_0 so that each matrix entry of $D(F_L - LQ_h)(X)$ has norm at most $K_0 \|X\|_\infty$. The corresponding remainder entries in physical coordinates are bounded by $K_0 h \|A\|_\infty / L$. With

$$D_\infty = \frac{m_L K_0}{L}, \qquad D_2 = K_0 \sqrt{m_L N_L}\, L^{d/2-1}, \qquad D_* = N_L K_0 L^{d-1}, \qquad S = \frac{1}{L-1}, \tag{45}$$

the physical transpose therefore satisfies

$$\left\| D\mathcal{R}_h(A)^T \right\|_{\infty\to\infty} \leq D_* h \|A\|_\infty . \tag{46}$$

Indeed the transpose converts an ordinary column sum to a physical row sum with factor $(Lh)^d/h^d = L^d$. This factor is already included in D_*.

Both Q_h and its physical transpose are bounded-field contractions. If $a = \|A\|_\infty$ belongs to the common nonlinear ball, the intermediate fields obey $\|A_t\|_\infty \leq 2a$. The spacings $h_t = L^{t-j}$ satisfy $\sum_{t<j} h_t \leq S$. Expanding the difference of the transposed nonlinear derivative product and its flat product thus proves

$$\left\| D\mathcal{R}(A)^T \right\|_{\infty\to\infty} \leq e^{2D_* a S} - 1 \leq c_* a \tag{47}$$

on a smaller fixed ball. The analogous derivative bounds in the bounded and physical Hilbert norms follow with D_∞ and D_2. These are absolute physical row or column estimates, not inferences from the fine dimension. The local analytic expansions also make $A \mapsto D\mathcal{R}(A)^T$ holomorphic in this operator norm. Cauchy's estimate on a slightly larger common ball gives

$$\left\| D\mathcal{R}(A)^T - D\mathcal{R}(A')^T \right\|_{\infty\to\infty} \leq c_{*,1} \|A - A'\|_\infty . \tag{48}$$

No bounded-field estimate for the gauge projection Π is used.

We also need a bound with two Hilbert directions, rather than just two bounded directions. Let K_2 bound the local bilinear Hessian $D^2 F_L$ in the maximum norms of its inputs; it bounds the full second

derivative, including its factor two. The scaling $D^2\mathcal{F}_h = (h/L)D^2F_L(hA)$, local Cauchy–Schwarz, and incidence counting give

$$\left|\left\langle w, D^2\mathcal{F}_h(A)[B,C]\right\rangle_{Lh}\right| \le k_2 h \left\|w\right\|_\infty \left\|B\right\|_{2,h} \left\|C\right\|_{2,h}, \qquad k_2 = K_2 N_L L^{d-1}. \tag{49}$$

For clarity, replace each local input maximum by its local Euclidean norm. Summing the product over output bonds and applying Cauchy–Schwarz costs at most N_L; restoring the fine measure then converts $(Lh)^d(h/L)$ to the factor in (49).

Let P_t be the derivative of the nonlinear prefix through stage t, and T_t the derivative of the suffix after stage $t+1$, both evaluated along the actual trajectory. The exact second-order chain rule is

$$D^2\mathcal{F}(A)[B,C] = \sum_{t<j} T_t D^2\mathcal{F}_{h_t}(A_t)[P_tB, P_tC]. \tag{50}$$

The two prefix Hilbert norms are each at most e^{2D_2aS} and the suffix transpose bounded-field norm at most e^{2D_*aS}. Pairing (50) with w and using (49) therefore proves the uniform form bound

$$\boxed{\left|\left\langle w, D^2\mathcal{F}(A)[B,C]\right\rangle_1\right| \le k_2 S e^{(4D_2+2D_*)aS} \left\|w\right\|_\infty \left\|B\right\|_{2,\eta} \left\|C\right\|_{2,\eta}.} \tag{51}$$

4.2 The exact reduced equation

Let $K = d^Td$, let $q = q_M$ be the Maxwell coarse precision, and let H be the flat minimizing extension in the matched source gauge. The flat identities are

$$QH = I, \qquad KH = Q^Tq, \qquad RD^TH = 0. \tag{52}$$

Set

$$\mathcal{X} = \ker Q \cap \ker(RD^T), \qquad \mathcal{C} = \Pi\Gamma\Pi^T = \Gamma - DZD^T, \qquad \Gamma = G_0 - H(q + a_{\mathrm{pen}})^{-1}H^T. \tag{53}$$

The established constrained-inverse identities give

$$Q\mathcal{C} = RD^T\mathcal{C} = \mathcal{C}Q^T = 0, \qquad \mathcal{C}K\xi = \xi \quad (\xi \in \mathcal{X}), \qquad \mathcal{C}KH = 0. \tag{54}$$

On a real finite regulator, $\mathcal{C}$ is the positive variational inverse of K on $\mathcal{X}$. Also $H^TK\xi = qQ\xi = 0$. The semidefinite q in these identities is not the positive auxiliary precision $q + a_{\mathrm{pen}}$.

For coarse B the exact local constraint chart is

$$A = \xi + H[B - \mathcal{R}(A)], \qquad \xi \in \mathcal{X}. \tag{55}$$

It gives $\mathcal{F}(A) = B$ and $RD^TA = 0$. Conversely, every small field satisfying these two conditions belongs to this chart, by taking $\xi = A - H[B - \mathcal{R}(A)]$. Its derivative in ξ is $T_A = (I + HD\mathcal{R}(A))^{-1}$ restricted to $\mathcal{X}$.

Write the ordinary-potential Wilson Euler derivative as

$$E(A) = KA + N(A), \qquad N(A) = S_{\mathrm{jet}}^T f_\eta(S_{\mathrm{jet}}A), \tag{56}$$

where S_{jet} collects the finitely many local values and ordinary first differences used by the plaquette jet. Stationarity of the action in the chart is equivalent to $\mathcal{C}T_A^TE(A) = 0$. The ordered inverse identity

$$T_A^T = I - D\mathcal{R}(A)^T(I + H^TD\mathcal{R}(A)^T)^{-1}H^T$$

and (52)–(55) transform this condition exactly into

$$\boxed{\xi = -\mathcal{C}N(A) + \mathcal{C}D\mathcal{R}(A)^T(I + H^TD\mathcal{R}(A)^T)^{-1}\{q[B - \mathcal{R}(A)] + H^TN(A)\}.} \tag{57}$$

In particular, the nonlinear constraint contributes the second term in (57); it cannot be omitted.

4.3 Uniform analytic contraction

Use the complete-field physical norm

$$\|A\|_{\mathcal{N}_\alpha} = \|A\|_\infty + \|\nabla A\|_\infty + [\nabla A]_\alpha. \tag{58}$$

It controls the complete C^α norms of the jet, uniformly in η. Telescoping controls differences of A at distances below one; its bounded norm handles larger distances. The flat extension and the constrained Green theorem proved above supply

$$H : \ell_c^\infty \longrightarrow \mathcal{N}_\alpha, \qquad \mathcal{C} : \ell_f^\infty \longrightarrow \mathcal{N}_\alpha, \tag{59}$$

$$\mathcal{C} S_{\text{jet}}^T : C_f^\alpha \longrightarrow \mathcal{N}_\alpha, \qquad q : \ell_c^\infty \longrightarrow \ell_c^\infty. \tag{60}$$

The equal exponent in (60) is essential. It uses the separate constrained biharmonic cancellation, including the fourth and fifth heat moments and the separation-dependent time split, not merely an estimate for G_0 or a theorem with loss of Hölder exponent.

The transpose estimates for H and its first differences give

$$H^T : \ell_f^\infty \longrightarrow \ell_c^\infty, \qquad H^T S_{\text{jet}}^T = (S_{\text{jet}} H)^T : C_f^\alpha \longrightarrow \ell_c^\infty. \tag{61}$$

For the second map the bounded input norm already suffices: the physical volume of a fine cell is fixed and the physical-cell Hilbert transpose rows of $S_{\text{jet}} H$ are summable. Thus (61) never differentiates a rough force coefficient.

The Wilson jet calculation supplies a holomorphic function f_η with $f_\eta(0) = Df_\eta(0) = 0$ and the uniform estimates

$$\|f_\eta(S_{\text{jet}} A)\|_{C^\alpha} \le c_f \|A\|_{\mathcal{N}_\alpha}^2 , \tag{62}$$

$$\|f_\eta(S_{\text{jet}} A) - f_\eta(S_{\text{jet}} A')\|_{C^\alpha} \le c_f (\|A\|_{\mathcal{N}_\alpha} + \|A'\|_{\mathcal{N}_\alpha}) \|A - A'\|_{\mathcal{N}_\alpha} . \tag{63}$$

These estimates concern the coefficient before the outside transpose in (56).

Shrink the common ball so that

$$\|H\|_{\infty\to\mathcal{N}_\alpha} \|D\mathcal{R}(A)\|_{\infty\to\infty} < \frac{1}{2}, \qquad \left\|H^T\right\|_{\infty\to\infty} \left\|D\mathcal{R}(A)^T\right\|_{\infty\to\infty} < \frac{1}{2}. \tag{64}$$

The first inequality makes (55) an analytic $\mathcal{N}_\alpha$ chart. Put $b = \|B\|_\infty$ and $h_N = \|H\|_{\infty\to\mathcal{N}_\alpha}$. Its contraction estimate gives

$$\|A(\xi, B)\|_{\mathcal{N}_\alpha} \le 2(\|\xi\|_{\mathcal{N}_\alpha} + h_N b), \qquad \|A(\xi, B) - A(\xi', B)\|_{\mathcal{N}_\alpha} \le 2 \|\xi - \xi'\|_{\mathcal{N}_\alpha} . \tag{65}$$

Define the coarse multiplier

$$m(A, B) = (I + H^T D\mathcal{R}(A)^T)^{-1} \{q[B - \mathcal{R}(A)] + H^T N(A)\}. \tag{66}$$

For $\|A\|_{\mathcal{N}_\alpha} , \|A'\|_{\mathcal{N}_\alpha} \le r$ in this chart, the preceding bounds imply

$$\|m(A, B)\|_\infty \le C(b + r^2), \qquad \|m(A, B) - m(A', B)\|_\infty \le C(b + r) \|A - A'\|_{\mathcal{N}_\alpha} . \tag{67}$$

For the first bound use inverse norm at most two, $\|\mathcal{R}(A)\|_\infty \le Cr^2$, and $H^T N = (S_{\text{jet}} H)^T f_\eta$. For the second use the inverse difference identity and (48); the change of the braces is $O(r \|A - A'\|)$, and the change of the inverse multiplies a quantity $O(b + r^2)$.

Let $\Phi_B(\xi)$ denote the right side of (57), with $A = A(\xi, B)$. Then

$$\|\Phi_B(\xi)\|_{\mathcal{N}_\alpha} \le C(r^2 + rb + r^3), \tag{68}$$

$$\|\Phi_B(\xi) - \Phi_B(\xi')\|_{\mathcal{N}_\alpha} \le C(b + r)\,\|\xi - \xi'\|_{\mathcal{N}_\alpha}\,. \tag{69}$$

To check the constraint contribution, expand the difference of $D\mathcal{R}(A)^T m(A, B)$ into its two factors and apply (47), (48), and (67); then use the bounded-input map $\mathcal{C}$ in (59). The force contribution uses (60) and (63). Both contributions lie in $\mathcal{X}$.

Choose a fixed small $\mathcal{X}$-ball and then $b_* > 0$ so that its chart image stays in the preceding balls, (68) maps it into itself, and (69) is below one half. This constructs a unique small solution $\xi(B)$ for every complex $\|B\|_\infty < b_*$. Uniform analytic contraction makes $\xi(B)$ and $A(B)$ holomorphic. The sharper bounds are

$$\|\xi(B)\|_{\mathcal{N}_\alpha} \le Cb^2, \qquad \|A(B) - HB\|_{\mathcal{N}_\alpha} \le Cb^2, \qquad \|A(B)\|_{\mathcal{N}_\alpha} \le Cb. \tag{70}$$

Indeed on $\|\xi\| \le Kb^2$, after reducing b_*, (65) gives $r \le 4h_N b$. The right side of (68) is then $C'b^2$. Take $K > 2C'$, shrink b_* again so that $Kb^2 \le h_N b$ and the Lipschitz bound is below one half, and use uniqueness in the original ball. The chart remainder $-H\mathcal{R}(A)$ proves the second estimate. If a complex coarse ball has distance δ from the boundary of the holomorphy ball and $\|A(B)\|_{\mathcal{N}_\alpha} \le C_0$ there, multivariable Cauchy gives, for $n \ge 1$,

$$\|D^n A(B)[B_1, \dots, B_n]\|_{\mathcal{N}_\alpha} \le C_0 n! \left(\frac{e}{\delta}\right)^n \prod_{i=1}^{n} \|B_i\|_\infty\,. \tag{71}$$

Use polyradii approaching δ/n and $n^n \le e^n n!$.

4.4 The real solution is a strict local minimum

For real fields the ordinary-coordinate Wilson Hessian has the form

$$D^2\mathcal{S}(A) = K + S_{\mathrm{jet}}^T W(A) S_{\mathrm{jet}}, \qquad \|W(A)\|_\infty \le C\,\|A\|_{\mathcal{N}_\alpha}\,. \tag{72}$$

This follows from the exact jet density whose quadratic Taylor part is the Maxwell density. It does not identify two different coordinate Hessians at a nonzero background.

The flat full inverse's first-derivative energy bound gives

$$\|z\|_{2,\eta}^2 + \|\nabla z\|_{2,\eta}^2 \le C_E \langle z, Kz\rangle_\eta \qquad (z \in \mathcal{X}). \tag{73}$$

In fact both the gauge and averaging penalties in the positive full precision vanish on $\mathcal{X}$. Apply the uniform bound on $(I, \nabla)G_0(I, \nabla)^T$ to its square-root factorization.

Let $z \in \mathcal{X}$ and $v = T_A z$. Differentiating the chart gives $v = z - HD\mathcal{R}(A)v$. The cross terms in its Maxwell energy vanish because $H^T K z = 0$, while q is positive semidefinite. Consequently

$$\langle v, Kv\rangle = \langle z, Kz\rangle + \langle D\mathcal{R}(A)v, qD\mathcal{R}(A)v\rangle \ge \langle z, Kz\rangle. \tag{74}$$

The uniform physical Hilbert bound $\|D\mathcal{R}(A)\|_{2\to 2} \le Cr$, together with the Hilbert bounds for H and ∇H, gives, after shrinking r,

$$\|v\|_2 \le 2\,\|z\|_2\,, \qquad \|v\|_2^2 + \|\nabla v\|_2^2 \le C\langle z, Kz\rangle. \tag{75}$$

No fine Hilbert-to-pointwise bound is used here.

Differentiate (55) once more. With $H_A = (I + HD\mathcal{R}(A))^{-1}H$ one obtains $A'' = -H_A D^2\mathcal{F}(A)[v,v]$. Moreover, $H_A^T E(A) = m(A,B)$ by (52) and (66). The exact reduced Hessian is therefore

$$\begin{aligned} D_\xi^2[\mathcal{S}(A(\xi,B))][z,z] &= \langle v, D^2\mathcal{S}(A)v\rangle \\ &\quad - \langle m(A,B), D^2\mathcal{F}(A)[v,v]\rangle. \end{aligned} \tag{76}$$

By (72), (74), and (75), the first term is at least $(1-Cr)\langle z, Kz\rangle$. Equations (51), (67), and (75) bound the second term in absolute value by $C(b+r^2)\langle z, Kz\rangle$. Thus throughout a sufficiently small real chart,

$$D_\xi^2[\mathcal{S}(A(\xi,B))][z,z] \geq \tfrac{1}{2}\langle z, Kz\rangle \geq c\,\|z\|_{2,\eta}^2\,. \tag{77}$$

Theorem 4.1 (Uniform local minimum on the fixed flat gauge slice)**.** *For sufficiently small real coarse B, the analytic field $A(B)$ constructed above is the unique stationary point in a fixed small chart of*

$$\mathcal{F}(A) = B, \qquad RD^T A = 0.$$

It is a strict local minimum of the normalized Wilson action on that chart. Its radius, the bounds (70)*, the analytic derivative estimates* (71)*, and the reduced Hessian bound* (77) *are independent of refinement and compatible periodic volume.*

Proof. The contraction supplies the unique small stationary field. The converse chart following (55) includes every small feasible field in the fixed gauge slice. Integrating the positive Hessian (77) along a line segment in the real ξ-chart proves strict local minimality and local uniqueness. □

4.5 Scope and normalization

The statement concerns the corrected fixed-reference constraint and a fixed flat gauge condition. A separate gauge-orbit argument is required to transfer it to an unrestricted constrained minimum or to a constraint formulated with ordinary block averages. It does not assert a minimum among large fields or a nonflat source gauge identification.

In dimension four the coupling-free action used here agrees with the deposited Wilson normalization: Raw I:464–479 puts g^{-2} outside the sum of $1 - \operatorname{Re}\tau(P)$, and I:12756–12760 specifies the physical trace pairing. The source left-exponential Hessian at a nonzero background is a different coordinate Hessian from the ordinary-potential Hessian used here; the latter has been derived directly from the same action.

On the infinite lattice the same uniformly local maps can define bounded stationary fields, but bounded fields generally have infinite total action. Without further finite-energy hypotheses, one must formulate stationarity and admissible local or decaying comparisons rather than a minimum of that total action. This module makes its minimum claim on finite periodic regulators.

5 Constraint-preserving gauges and coverage of the flat slice

This module concerns finite compatible periodic regulators, fixed odd blocking ratio L, refinement $M = L^j$, fine spacing $\eta = M^{-1}$, and final spacing one. Fix $0 < \alpha < 1$. All constants below are independent of refinement and volume. We use the corrected nonlinear map with fixed identity reference, $\mathcal{F}(A) = \mathcal{F}^{(j)}(A)$, and its ordinary potential differential $Q_A = D\mathcal{F}(A)$. This is not renamed as the separately based source $Q(U)$. The flat scalar projection denoted R earlier is denoted R_0 here. The potential norms N_α, $\mathcal{N}_\alpha$ and $\mathcal{X}_\alpha$ refer to the same complete physical norm, up to the fixed collar equivalences stated below.

Let $\mathcal{X}_\alpha$ be the complete-field norm consisting of the bounded potential, its bounded physical first differences and the local α-Hölder seminorms of those differences. Let $\mathcal{Y}_\alpha$ be the scalar C^α norm,

including the supremum, and $\mathcal{Z}_\alpha$ the scalar $C^{2,\alpha}$ norm, including all physical differences through order two and the α-Hölder members at order two. Fixed collar enlargements give equivalent norms. At fixed refinement these are Banach spaces. The estimates below are uniform in that refinement. For distances above a fixed collar size, supremum norms supply the corresponding global increments. No field is sharply cut off before differentiating.

The purpose is to upgrade a local minimum on the fixed flat gauge slice to a local minimum of the same nonlinear constraint, modulo constraint-preserving gauge transformations. This requires a new gauge argument; strict minimality on the slice alone is insufficient.

5.1 The induced coarse gauge of the corrected average

For the ordinary group average M and one-step endpoint mean $a_y(U)$, use the exact identities already checked in Notebook 14:

$$C(U)(c) = a_{c_-}(U)^{-1} M(U)(c) a_{c_+}(U), \qquad M(U^g) = M(U)^{g|_{\text{coarse}}},$$

where $U^g(e) = g(e_-)U(e)g(e_+)^{-1}$. Direct multiplication gives

$$C(U^g) = C(U)^{k(U,g)}, \qquad k_y(U,g) = a_y(U^g)^{-1} g(y) a_y(U). \tag{78}$$

All factors are on the common small-field branch. The equality is exact, not a linearization. In particular $k(U,1) = 1$.

To compute its differential, set

$$P_x = U(\Gamma_{yx}) = e^{iY_x}, \quad \overline{Y}_y = \sum_x L^{-d} Y_x, \quad a_y(U) = e^{i\overline{Y}_y}, \qquad \mathcal{J}_X Z = \int_0^1 \mathrm{Ad}_{e^{itX}}\, Z\, dt.$$

For $g(s) = e^{is\lambda}$ the contour left increment is $\lambda(y) - \mathrm{Ad}_{P_x}\lambda(x)$. The logarithm and exponential differentials therefore give the following explicit coarse gauge operator:

$$T_U\lambda(y) = \mathrm{Ad}_{a_y(U)^{-1}}\left[\lambda(y) - \mathcal{J}_{\overline{Y}_y} \sum_x L^{-d} \mathcal{J}_{Y_x}^{-1}(\lambda(y) - \mathrm{Ad}_{P_x}\lambda(x))\right]. \tag{79}$$

Thus $(1/i)\partial_s k_y(U, e^{is\lambda})|_0 = T_U\lambda(y)$. The gauge parameter has angle units; no factor h is inserted into $g = e^{i\lambda}$. At $U = 1$ this formula reduces exactly to the normalized scalar block average $Q_{0,h}$.

For $U = e^{ihA}$, every coefficient in (79) is a finite local analytic function of hA. Its difference from its value at zero is $O(h\,\|A\|_\infty)$. Finite one-step row incidence and physical-adjoint column incidence then give

$$\|T_U - Q_{0,h}\|_{\text{av}} \le C_L h\,\|A\|_\infty, \qquad \|T_U\|_{\text{av}} \le 1 + C_L h\,\|A\|_\infty. \tag{80}$$

Here the absolute Schur norm uses physical measures and includes the bounded-field norm of the transpose. Its column factor is the fixed volume ratio L^d. An evaluation at the block center in (79) consequently costs a fixed L constant, not a power of the full refinement. The flat scalar averages and their physical adjoints have row sums one.

Let C_t be the corrected iterates of $U = e^{i\eta A}$, and let $g_0 = g$, $g_{t+1} = k(C_t, g_t)$. Induction in (78) gives the exact multilevel gauge law $C^{(j)}(U^g) = C^{(j)}(U)^{g_j}$. Differentiating at $g = 1$ gives the scalar operator

$$\mathcal{G}_A = T_{C_{j-1}} \cdots T_{C_0}, \qquad \mathcal{G}_0 = Q_0. \tag{81}$$

These backgrounds are the corrected iterates, unlike the ordinary reference trajectory used for the source $Q(U)$. The proven bound $C_t = e^{ih_t A_t}$, $\|A_t\|_\infty \le 2\,\|A\|_\infty$, and $\sum_{t<j} h_t \le 1/(L-1)$ imply

$$\|\mathcal{G}_A - Q_0\|_{\text{av}} \le e^{C\|A\|_\infty/(L-1)} - 1. \tag{82}$$

Its coefficients have fixed final physical range. The same estimate, up to a fixed constant, holds in weighted physical-cell norms. The operator is holomorphic in the common bounded-field ball; local analytic coefficients and finite range first give holomorphy at each refinement, and the displayed uniform bounds give uniform Cauchy bounds on smaller balls.

Define the ordinary-coordinate fine gauge generator and its coarse analogue by

$$X_A = \mathcal{J}_{\eta A}^{-1} D_U, \qquad U = e^{i\eta A}, \quad D_U\lambda(e) = \eta^{-1}(\mathrm{Ad}_{U_e}\,\lambda(e_+) - \lambda(e_-)). \tag{83}$$

A gauge $e^{is\lambda}$ has ordinary potential velocity $-X_A\lambda$. At the output, with $B = \mathcal{F}(A)$ and $V = e^{iB}$, differentiation of the multilevel group law yields

$$Q_A X_A = \mathcal{J}_B^{-1} D_V \mathcal{G}_A. \tag{84}$$

The output spacing is one. In particular $X_A\lambda$ is tangent to the nonlinear constraint whenever $\mathcal{G}_A\lambda = 0$.

5.2 Smooth scalar tools for the flat gauge

Retain $\mathcal{L} = D^T D$, the scalar flat projection R_0, and Z of the previous flat notebooks. Put

$$T = G' R_0 = Z\mathcal{L}. \tag{85}$$

The exact identities give

$$Q_0 T = 0, \qquad R_0 \mathcal{L} T = R_0. \tag{86}$$

The following scalar estimates are needed:

$$\|R_0 f\|_{\mathcal{Y}_\alpha} \le C\,\|f\|_{\mathcal{Y}_\alpha}\,, \qquad \|Tf\|_{\mathcal{Z}_\alpha} \le C\,\|f\|_{\mathcal{Y}_\alpha}\,. \tag{87}$$

Here is a direct justification from the already proved canceled aliases. With $A_0 = a_0 a_0^-$ and $D_{s,2} = A_0 + P_0^2 \sum_{r\neq 0} a_r a_r^-/P_r^2$, the high block of $T = Z\mathcal{L}$ is

$$T_{rs} = \frac{\delta_{rs}}{P_r} - \frac{P_0^2 a_r^- a_s}{P_r^2 P_s D_{s,2}}, \qquad r, s \neq 0.$$

Its mixed entries are $T_{0s} = -a_0^- a_s/(P_s D_{s,2})$ and $T_{r0} = -P_0 a_r^- a_0/(P_r^2 D_{s,2})$, while $T_{00} = P_0 R_{s,2}/D_{s,2}$. Subtract $(\mathcal{L}+1)^{-1}$. The high diagonal remainder has heat factor $1 - e^{-t}$, making its two-output-difference α-Hölder estimate integrable. All high rank output vectors are a_r^-/P_r^2, which are absolutely summable after $2+\alpha$ differences; input rows are in alias ℓ^2. The scalar massive part maps C^α to $C^{2,\alpha}$ by zero-row heat cancellation and the time split at the square of the output separation. This is the same second/third heat derivative calculation used for the right-divergence Schauder theorem, now with both derivatives on the output.

Also $R_0 = I - P_0^{\mathrm{gauge}}$, where the rank-one Bloch projection has output vector

$$\chi_0 = a_0^-, \qquad \chi_r = P_0 a_r^-/P_r \quad (r \neq 0), \qquad P_0^{\mathrm{gauge}} = \chi\chi^-/D_{s,2}.$$

It maps bounded fine fields to $C^{1,\beta}$ for every fixed $\beta < 1$, by absolute output alias sums and input ℓ^2. This proves the first bound in (87) for fixed $\alpha < 1$. Periodic reconstruction and contour shifting can be applied to these complete operators; hence the same estimates have exponential cell locality and weighted versions. These arguments do not claim a second-difference estimate on arbitrary bounded forcing for the massive inverse.

We also require a sufficiently smooth scalar right inverse of Q_0. Define H_b by the canceled scalar biharmonic symbols

$$(H_b)_0 = \frac{a_0^-}{D_{s,2}}, \qquad (H_b)_r = \frac{P_0^2 a_r^-}{P_r^2 D_{s,2}} \quad (r \neq 0). \tag{88}$$

Then

$$Q_0 H_b = I, \qquad \|H_b c\|_{\mathcal{Z}_\alpha} \le C \|c\|_\infty . \tag{89}$$

On nonzero real fibers this is exactly $\mathcal{L}^{-2} Q_0^T (Q_0 \mathcal{L}^{-2} Q_0^T)^{-1}$. The canceled formula supplies the constant-mode continuation, so the complete operator has the periodic gluing needed above. The identity follows by multiplying the scalar row a_r into (88). The bound follows from the same a_r/P_r^2 absolute alias sums; these even allow $3+\beta$ differences for $\beta < 1$. Central entries are bounded and analytic through $P_0 = 0$. This is a separately constructed scalar extension, not the less smooth extension $G'^2 Q_0^T (Q_0 G'^2 Q_0^T)^{-1}$.

5.3 An invertible gauge-fixing differential

Set $F_A = \mathcal{G}_A - Q_0$. On a sufficiently small field ball,

$$T_A^s = (I + H_b F_A)^{-1} \tag{90}$$

is a uniformly bounded holomorphic operator on $\mathcal{Z}_\alpha$. Indeed $F_A : \mathcal{Z}_\alpha \to \ell^\infty$ has norm $O(\|A\|_\infty)$ and H_b satisfies (89). The inverse differs from identity by the same small order. If $Q_0 \psi = 0$ and $\lambda = T_A^s \psi$, applying Q_0 to $\lambda + H_b F_A \lambda = \psi$ gives

$$\mathcal{G}_A \lambda = 0. \tag{91}$$

There is a uniform estimate

$$\|(X_A - D)\lambda\|_{\mathcal{X}_\alpha} \le C \|A\|_{\mathcal{X}_\alpha} \|\lambda\|_{\mathcal{Z}_\alpha} . \tag{92}$$

To see the physical scaling, write $D_U = D + E_s(A)$ with $E_s(A)\lambda = \eta^{-1}(\mathrm{Ad}_{e^{i\eta A}} - I)\lambda(\cdot + \eta e_\mu)$. The coefficient in this expression extends analytically to $\eta = 0$ and has $C^{1,\alpha}$ norm at most $C\|A\|_{\mathcal{X}_\alpha}$. Also $\mathcal{J}_{\eta A}^{-1} - I$ has that norm at most $C\eta \|A\|_{\mathcal{X}_\alpha}$. Discrete product and shift estimates then prove (92), using two derivatives of λ. Every derivative of a coefficient is paid by a derivative of A; there is no inverse-mesh loss. These maps are holomorphic in their indicated Banach norms on a common smaller ball.

The operator $R_0 D^T : \mathcal{X}_\alpha \to \mathcal{Y}_\alpha$ is uniformly bounded. Let $\mathcal{R}_\alpha = \{f \in \mathcal{Y}_\alpha : R_0 f = f\}$, a closed subspace because of (87). Define on it

$$\mathcal{M}_A = R_0 D^T X_A T_A^s T. \tag{93}$$

By (86), $\mathcal{M}_0 = I$ on $\mathcal{R}_\alpha$. The preceding estimates give

$$\|\mathcal{M}_A - I\|_{\mathcal{R}_\alpha \to \mathcal{R}_\alpha} \le C \|A\|_{\mathcal{X}_\alpha} . \tag{94}$$

Its range is in $\mathcal{R}_\alpha$ because it begins with R_0. For a fixed small radius its Neumann inverse is uniformly bounded and holomorphic. Define the scalar correction operator

$$\mathcal{L}_A = T_A^s T \mathcal{M}_A^{-1} R_0 D^T. \tag{95}$$

It satisfies the exact identities

$$\mathcal{G}_A \mathcal{L}_A v = 0, \qquad R_0 D^T X_A \mathcal{L}_A v = R_0 D^T v, \tag{96}$$

and the stronger norm estimate

$$\|\mathcal{L}_A v\|_{\mathcal{Z}_\alpha} \le C \left\| R_0 D^T v \right\|_{\mathcal{Y}_\alpha} . \tag{97}$$

Consequently any constrained tangent $Q_A v = 0$ decomposes as

$$v = (v - X_A \mathcal{L}_A v) + X_A \mathcal{L}_A v, \tag{98}$$

where the first term is both constrained and in the fixed flat gauge, and the second is a constraint-preserving gauge tangent. The first claim uses (84) and (96). Its tangent space is exactly the image of $X = \ker Q \cap \ker R_0 D^T$ under the constraint-chart derivative $T_A = (I + HD\mathcal{R}(A))^{-1}$: for a constrained fixed-gauge tangent v, the vector $z = (I + HD\mathcal{R}(A))v$ satisfies $Qz = 0$ and $R_0 D^T z = 0$. A stationary Wilson field on the fixed slice is therefore stationary on the full constraint: the action derivative vanishes on the first term by slice stationarity and on the second by exact gauge invariance.

5.4 A constraint-preserving flow into the slice

For real fields use the differential equation

$$\frac{dA(t)}{dt} = -X_{A(t)}\mathcal{L}_{A(t)}A(t), \qquad A(0) = A^{\mathrm{in}}. \tag{99}$$

The vector field is locally Lipschitz on the small $\mathcal{X}_\alpha$ ball, uniformly in refinement. Its negative sign matches the actual gauge velocity in (83). Along any solution in that ball,

$$R_0 D^T A(t) = e^{-t} R_0 D^T A^{\mathrm{in}}, \qquad \mathcal{F}(A(t)) = \mathcal{F}(A^{\mathrm{in}}), \qquad \mathcal{S}_\eta(e^{i\eta A(t)}) = \mathcal{S}_\eta(e^{i\eta A^{\mathrm{in}}}). \tag{100}$$

The first identity follows directly from (96); the second from (84); the third from gauge invariance. All parameter operators preserve real fields. Thus the flow integrates unitary gauge transformations on the finite regulator.

By (97) and the uniform bound for X_A,

$$\|A'(t)\|_{\mathcal{X}_\alpha} \le Ce^{-t} \left\|R_0 D^T A^{\mathrm{in}}\right\|_{\mathcal{Y}_\alpha}. \tag{101}$$

Choose the initial ball sufficiently small that its radius plus this total displacement lies within the established operator domain. Standard continuation then keeps the solution in that domain for every $t \ge 0$. The integrable speed proves convergence in $\mathcal{X}_\alpha$ to $A^\flat$, with

$$R_0 D^T A^\flat = 0, \quad \mathcal{F}(A^\flat) = \mathcal{F}(A^{\mathrm{in}}), \quad \mathcal{S}_\eta(e^{i\eta A^\flat}) = \mathcal{S}_\eta(e^{i\eta A^{\mathrm{in}}}), \quad \left\|A^\flat - A^{\mathrm{in}}\right\|_{\mathcal{X}_\alpha} \le C \left\|R_0 D^T A^{\mathrm{in}}\right\|_{\mathcal{Y}_\alpha}. \tag{102}$$

The gauge parameter is integrable even in $\mathcal{Z}_\alpha$, so the gauge transformations themselves converge uniformly to a unitary limit. Hence the limiting field remains in the original gauge orbit, not merely at the same action value.

For a fixed slice field A_* the same construction can be made in a neighborhood of A_*. Since $R_0 D^T A_* = 0$, the total displacement is bounded by $C\|A^{\mathrm{in}} - A_*\|_{\mathcal{X}_\alpha}$. Choosing this neighborhood inside the strict slice-minimality neighborhood ensures that $A^\flat$ belongs to it. Therefore, if A_* is a strict local minimum on

$$\mathcal{F}(A) = B, \qquad R_0 D^T A = 0,$$

then it is a local minimum of $\mathcal{F}(A) = B$ without imposing the flat gauge condition. Equality in a sufficiently small neighborhood implies $A^\flat = A_*$ and hence that the initial field is gauge equivalent to A_*. This is local minimality modulo the constraint-preserving gauge orbit, not an absolute global minimum among all lattice configurations.

The argument supplies uniform local gauge coverage for the stated global small-field class. It does not cover arbitrary plaquette-small gauge orbits, changing retained domains, topologically nontrivial bundles, or a continuum quantum measure. No finite total action is assigned to arbitrary bounded infinite-lattice fields here.

6 Transfer to the ordinary homogeneous averaging constraint

The preceding minimizer was constructed for the corrected fixed-reference constraint $C^{(j)}(U) = D = e^{iB}$. The ordinary constraint is $M^{(j)}(U) = D$, with $M^{(j)}$ the homogeneous iteration of the ordinary group average. These two equalities are distinct. This section proves a transfer between their small field minimizers using actual gauge transformations. All fields are on finite compatible periodic regulators; final spacing is one. The radii and constants below are independent of refinement and volume. The bare Wilson action is gauge invariant.

6.1 An exact multilevel gauge identity

Let $C_t = C^{(t)}(U)$ and $b_0(U) = 1$. Notebook 14 proves

$$b_{t+1}(U) = b_t(U)|_{t+1}\, a(C_t), \qquad M^{(t)}(U) = C_t^{\,b_t(U)}. \tag{103}$$

Here restriction is to the next coarse sites, and $a(C_t)$ is the endpoint mean at that step. In particular, on a fixed sufficiently small bounded-potential ball,

$$\|b_j(U) - 1\|_\infty + \left\|b_j(U)^{-1} - 1\right\|_\infty \le C\,\|A\|_\infty\,, \qquad U = e^{i\eta A}. \tag{104}$$

These group-valued maps are holomorphic with uniform bounds on a common complex ball. This follows either directly from the product in (103) and the summability of the physical spacings, or from the established corrected trajectory bounds. Only a coarse supremum bound is needed here.

For a fine gauge g, let $k_0 = g$ and let k_t be its induced gauge under the corrected iteration, so that $C^{(t)}(U^g) = C_t^{k_t}$. The one-step formula gives

$$k_{t+1} = a(C_t^{k_t})^{-1}\, k_t|_{t+1}\, a(C_t).$$

Multiplication with (103), applied also to U^g, proves by induction

$$b_t(U^g)k_t = g|_t\, b_t(U), \qquad k_j(U,g) = b_j(U^g)^{-1}\, g|_j\, b_j(U). \tag{105}$$

Indeed the adjacent factors $a(C_t^{k_t})$ and their inverses cancel before the inductive identity is used. No multiplicative cocycle for the logarithmic endpoint average is assumed.

6.2 Uniform analytic action of smooth fine gauges

Write $\mathcal{X}_\alpha = C^{1,\alpha}$ for the complete potential norm and $\mathcal{Z}_\alpha = C^{2,\alpha}$ for scalar gauge parameters, as in the preceding section. There is a uniform analytic map

$$\mathscr{A}_\eta(A,\lambda)(e) = \frac{1}{i\eta}\log\!\left[e^{i\lambda(e_-)}e^{i\eta A(e)}e^{-i\lambda(e_+)}\right] \tag{106}$$

from a fixed small ball in $\mathcal{X}_\alpha \times \mathcal{Z}_\alpha$ to $\mathcal{X}_\alpha$, and

$$\|\mathscr{A}_\eta(A,\lambda)\|_{\mathcal{X}_\alpha} \le C\big(\|A\|_{\mathcal{X}_\alpha} + \|\lambda\|_{\mathcal{Z}_\alpha}\big). \tag{107}$$

To justify the apparent division by η, put $\ell = \lambda(e_-)$ and $v = \nabla_\mu\lambda(e_-)$. The group element in (106) equals

$$e^{i\eta\,\mathrm{Ad}_{e^{i\ell}}A(e)}e^{i\ell}e^{-i(\ell+\eta v)}.$$

For every small (A,ℓ,v) this analytic product is one at $\eta = 0$. Its logarithm is therefore divisible by η as an analytic function, with a uniform extension for $0 \le \eta \le 1$. All its finite-dimensional derivatives are uniform on a smaller fixed complex jet ball. Discrete product, shift and analytic-composition estimates applied to $(A,\lambda,\nabla\lambda)$ prove (107) and holomorphy in the indicated Banach norms. A first potential difference uses at most two scalar differences. The same argument applies to its α-Hölder seminorm, so no inverse-mesh loss occurs. For real Hermitian traceless arguments the output remains Hermitian traceless on the common logarithm branch.

6.3 Prescribing the induced coarse gauge

Use the smooth block-mean right inverse H_b constructed in (88)–(89). It is a scalar operator tensored with the gauge Lie algebra. For bounded coarse θ define

$$\mathcal{K}(A,\theta) = \frac{1}{i}\log k_j(e^{i\eta A}, e^{iH_b\theta}). \tag{108}$$

The exact formula (105), the bound (104), the gauge-action estimate and $H_b : \ell^\infty \to \mathcal{Z}_\alpha$ show that $\mathcal{K}$ is uniformly bounded and holomorphic on a fixed ball in $\mathcal{X}_\alpha \times \ell^\infty$. Evaluation of $H_b\theta$ at coarse sites is bounded by its supremum norm. Thus this step does not presume that H_b interpolates point values.

At $\theta = 0$ the induced gauge is one, so $\mathcal{K}(A,0) = 0$. At the joint origin its differential is

$$D_\theta\mathcal{K}(0,0) = \mathcal{G}_0 H_b = Q_0 H_b = I. \tag{109}$$

Uniform Cauchy estimates on a smaller ball consequently give

$$\|D_\theta\mathcal{K}(A,\theta) - I\|_{\infty\to\infty} \le C(\|A\|_{\mathcal{X}_\alpha} + \|\theta\|_\infty). \tag{110}$$

For small A and any sufficiently small bounded coarse ω, the contraction

$$\theta = \omega - [\mathcal{K}(A,\theta) - \theta]$$

therefore has a unique solution in a fixed small ball. The solution satisfies $\|\theta\|_\infty \le 2\|\omega\|_\infty$ and depends holomorphically on (A,ω), with uniform Cauchy bounds on smaller balls. It is real for real inputs. Thus any small prescribed coarse gauge $e^{i\omega}$ is realized as the induced corrected gauge of a fine gauge $e^{iH_b\theta}$ of uniformly controlled smoothness.

Suppose now that $U = e^{i\eta A}$ satisfies $M^{(j)}(U) = D$. By (103), $C^{(j)}(U)^{b_j(U)} = D$. Prescribe

$$\omega = \frac{1}{i}\log b_j(U), \qquad k_j(U,g) = b_j(U).$$

The preceding construction produces a fine gauge g such that

$$C^{(j)}(U^g) = D, \qquad \mathcal{S}_\eta(U^g) = \mathcal{S}_\eta(U), \qquad \|\mathscr{A}_\eta(A, \log(g)/i)\|_{\mathcal{X}_\alpha} \le C\|A\|_{\mathcal{X}_\alpha}. \tag{111}$$

Here $\log(g)/i = H_b\theta$ on the chosen branch. This map takes sufficiently small ordinary-feasible fields into uniformly small corrected-feasible fields.

6.4 An ordinary-feasible representative of the minimizer

Let $U_*(B) = e^{i\eta A_*(B)}$ be the corrected minimizer constructed above, so $C^{(j)}(U_*) = D = e^{iB}$ and $\|A_*(B)\|_{\mathcal{X}_\alpha} \le C\|B\|_\infty$. We need a smooth fine gauge with specified *point values* at the final coarse sites. For this purpose construct a different operator E_{pt} as follows. Choose a fixed smooth scalar bump φ supported in $(-1/2,1/2)^d$ and satisfying $\varphi(0) = 1$. On the periodic coarse lattice put

$$(E_{\mathrm{pt}}\omega)(x) = \sum_{n\in\mathbb{Z}^d} \varphi(x-n)\omega(n), \tag{112}$$

with the coefficients extended periodically; evaluate the result at fine sites. This is well defined on every compatible torus, including the smallest coarse periods. Its derivatives through order three are bounded by $C\|\omega\|_\infty$, since the translates have uniformly bounded overlap. Sampling and the integral formula for finite differences give, uniformly in η,

$$E_{\mathrm{pt}}\omega|_{\mathrm{coarse}} = \omega, \qquad \|E_{\mathrm{pt}}\omega\|_{\mathcal{Z}_\alpha} \le C\|\omega\|_\infty. \tag{113}$$

Set

$$\omega_*(B) = \frac{1}{i}\log b_j(U_*(B))^{-1}, \quad h_*(B) = e^{iE_{\rm pt}\omega_*(B)}, \quad U_{\rm ord}(B) = U_*(B)^{h_*(B)}. \tag{114}$$

Ordinary gauge covariance and $h_*|_{\rm coarse} = b_j(U_*)^{-1}$ give

$$M^{(j)}(U_{\rm ord}(B)) = D, \qquad \mathcal{S}_\eta(U_{\rm ord}(B)) = \mathcal{S}_\eta(U_*(B)). \tag{115}$$

Indeed with the convention $V^g = g_- V g_+^{-1}$, $(V^b)^h = V^{hb}$, so the two coarse gauges cancel in the required order. The ordinary potential $A_{\rm ord}(B) = \log U_{\rm ord}(B)/(i\eta)$ is holomorphic on a uniform coarse supremum ball and obeys

$$\|A_{\rm ord}(B)\|_{\mathcal{X}_\alpha} \le C\,\|B\|_\infty\,. \tag{116}$$

Theorem 6.1 (Ordinary homogeneous constrained minimum)**.** *Fix $0 < \alpha < 1$, a fixed odd blocking ratio $L \ge 3$, and the compact matrix gauge group and trace normalization used in the Wilson calculation. There are $b_0, r_0 > 0$, independent of the homogeneous refinement $M = L^j$ and finite compatible periodic volume, with the following property. For every real coarse potential $\|B\|_\infty < b_0$, the field $U_{\rm ord}(B)$ in (114) satisfies the ordinary constraint $M^{(j)}(U) = e^{iB}$ and minimizes the bare Wilson action among all fields $U = e^{i\eta A}$ satisfying that constraint and $\|A\|_{\mathcal{X}_\alpha} < r_0$. Equality in this small field class implies gauge equivalence to $U_{\rm ord}(B)$. Its ordinary potential has the uniform bound (116) and a holomorphic extension to a fixed complex coarse-data ball.*

Proof. The reduced-Hessian proof gives strict convexity of the corrected fixed-slice action on a fixed ball of the linear coordinate $\xi \in X$, for all sufficiently small B. In particular the corrected minimizer is the unique minimum on that ball. The constraint chart covers every sufficiently small corrected-feasible fixed-slice field, and $\xi = (I - HQ)A$ there; this coordinate is uniformly small in $\mathcal{X}_\alpha$ since Q is bounded into coarse supremum norm and H maps that norm into $\mathcal{X}_\alpha$.

Given an ordinary-feasible field with $\|A\|_{\mathcal{X}_\alpha} < r_0$, first apply (111), and then the constraint-preserving flow into the flat slice. Both operations preserve the action and enlarge the potential norm by at most a fixed factor. Choose r_0 so small that the resulting fixed-slice field belongs to the chart and its ξ belongs to the strict-convexity ball. The action comparison with $U_*(B)$ now follows from that convexity. Choose b_0 smaller, if necessary, so that $\|A_{\rm ord}(B)\|_{\mathcal{X}_\alpha} < r_0$ and the minimizer coordinate belongs to the same convex ball. Equation (115) transfers the comparison to $U_{\rm ord}(B)$. All radii can be chosen with fixed strict margins and are uniform in refinement and volume.

If equality holds, strict convexity identifies the final fixed-slice field with $U_*(B)$. Both earlier operations are genuine gauge transformations, so the original field is gauge equivalent to $U_{\rm ord}(B)$. Since both have the same ordinary coarse constraint, the restriction of a connecting gauge to the coarse sites stabilizes that coarse field. No uniqueness of the gauge parameter is claimed. The analyticity and norm statements follow from (104), (107) and the analytic corrected minimizer. □

This is a minimum in a fixed small *global potential* class on each finite homogeneous regulator. It is not a global minimum over all lattice gauge fields, nor a claim about arbitrary local-frame or mixed-history constraints. The source's notation $M_{\mathfrak{B}}(U) = D$ includes more general geometries; only the explicitly defined ordinary homogeneous iteration is proved here. No continuum quantum measure or physical mass gap follows from this theorem.

7 What this supplies to the construction program

The new result supplies an actual nonlinear classical background with an exact ordinary coarse constraint. The reduced stationary equation includes both the nonlinear Wilson force and the full curvature of the constraint manifold. It has uniform analytic solutions, and the real action comparison survives removal

of the fixed gauge and transfer to ordinary averages. In this homogeneous small-global-field class, the minimizer is no longer merely an input that must be assumed.

Notebook 16 now derives a uniform local response and a controlled expansion of the coarse classical action from these inputs. Its weighted argument includes the derivative of the transposed constraint map and the coarse multiplier. It proves factorial responses with one weighted direction at every order and controls the particular ordinary representative through its explicit gauge formula. A fixed-exponent connected-tree bound for multiple insertions is stronger and remains unproved.

Even a local classical coarse action would still be only part of the required quantum renormalization step. Integration of fluctuations brings determinants, gauge-coordinate Jacobians, overlapping interactions and regions outside this global chart. Those contributions must be bounded uniformly and iterated with the coupling before a continuum probability law can be claimed. None is replaced by the minimization theorem.

References

[1] E. Sussman and collaborators, *Yang–Mills-LLM-experiment*, deposited raw accounts I–III at commit `e392ef45013a581664bab76b90867170b30fe10e`; repository. Raw I:454–479 gives the Wilson normalization, I:12756–12760 the physical bilinear pairing, I:15471–15478 and I:15643–15648 the source quadratic left-exponential chart, and I:43103–43107 the bare constrained action. The ordinary constraint is displayed at I:43098. Original journal papers are not freshly certified by these deposited matches.

[2] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 8–14 in the present workspace, 16 October 2026. Flat source-gauge extension, current and regularity estimates; nonlinear constraint chart; nonflat reference comparison and Wilson energy stability.